Solution

4 -8 to +7

-2 3 to 2 3 -1

Code No: 20CS340 USN:

JSS Mahavidyapeetha

JSS Science And Technology University

(Established Under JSS Science and Technology University Act No. 43 of 2013)
(Formerly Known as SJCE)

JSS SCIENCE AND TECHNOLOGY UNIVERSITY MYSURU

Branch: CS&E
Class: IIISemester

Test - 1

Section: A,B & C

Computer Organization

Duration: 1 Hour
Date: 15/12/2021

Time: 9.30AM-10.30AM

Marks: 20
Day: Wednesday

Course Outcomes:

CO-1	Comprehend and analyze the basic structure, operation of computers and the perspective of Computer architect.
CO-2	Illustrate the concept of programs as sequences of machine instructions.
CO-3	Demonstrate functioning of different sub systems, such as processor, Input/Output and memory.
CO-4	Analyze and design simple arithmetic, logical and processing units.

Note: Answer any TWO full questions:

Q. No.	CO/Cognitive Domain	Questions	Marks
1	CO1/Remember and Analyze	a) With the help of block a diagram that shows the connection between the processor and main memory, list the steps needed to execute machine instruction STORE R2, LOC.	05
		b) Represent the decimal values 19 and -24, as signed, 7- bit numbers in the following binary formats: i) Sign magnitude ii) 1's complement iii) 2's complement Also perform 2's complement binary addition and subtraction between the two and check whether or not overflow occur.	05
2	CO1/Remember and Analyze	a) Explain the following terms. Use diagrams/examples wherever necessary. i) Assembler directives ii) Word alignment iii) Big-Endian and Little-Endian memory assignments	05
		b) Registers R1 and R2 of a computer contain the decimal values 1500 and 5500. Name the type of addressing mode and calculate EA of memory operand in each of the following instructions? i) LOAD R5, (R1) ii) MOVE R5, 50(R2) iii) STORE R5, 30(R1,R2) iv) ADD R5, –(R2) v) SUB R5, (R1)+	05
3	CO2/Remember and Apply	a) With illustrations, discuss Logical and arithmetic shift instructions. Also give assembly language instructions to do the following tasks on a 16-bit register: i) Clear the lower order 4 bits (bit0-bit3) of register R2 ii) set the most significant bit of register R2. Both the operations should not change the remaining bits.	05
		b) With a neat diagram explain the stack data structure. Also implement PUSH and POP operations for RISC and CISC machines separately. Assume a byte-addressable memory with a 32-bit word length.	05

SCHEME - TEST-1 CO

1 a) Diagram ——— (02 M)

Listing steps to execute STORE R2, LOC

Fetch instrn (01M) — Writing R2 into LOC (02M)

b)

	Sign-magnitude (64 32 16 8 4 2 1)	1's	2's
19 →	0 0 1 0 0 1 1	0 0 1 0 0 1 1	0 0 1 0 0 1 1
(02M) 24 →	0 0 1 1 0 0 0	1's → 1 1 0 0 1 1 1	2's → 1 1 0 1 0 0 0
−24 →	1 0 1 1 0 0 0		

Add 19 + −24

```
  0 0 0 1 0 0 1 1   (19)
+ 1 1 1 0 1 0 0 0   (-24)
-----------------
  1 1 1 1 1 0 1 1   (-5)
```

2's → 0 0 0 0 0 1 0 1 (-5) (3M)

range -2^{n-1} to $2^{n-1}-1$

-2^6 to 2^6-1

−64 to +63

∴ no overflow in both the cases.

sub 19 − (−24)

19 + 24

```
  0 0 0 1 0 0 1 1   (19)
+ 0 0 0 1 1 0 0 0   (24)
-----------------
  0 0 1 0 1 0 1 1   (43)
```

32 8 2 1

2 a) Definition of

(i) Assembler directives + Eg. — (1½ M)

(ii) Word alignment + Eg — (1½ M)

(iii) Big & Little Endian + Dgm. — (03 M)

b) (i) Load R5, (R1)

(ii) Move R5, 50(R2)

(iii) Store R5, 30 (R1, R2)

(iv) Add R5, -(R2)

(v) Sub R5, (R1)+

Addressing mode	EA
Reg. Indirect	1500
Base Index	50+5500
Reg. Base Index	30+1500+5500
Auto decrement	5500-4
Auto increment	EA = 1500, later 1500+4

— 1 X 5 = 5 M

3 a) Logical shift Left — LShiftL R2, R2, #1 + dgm (Eg)

Logical shift Right — LShiftR R2, R2, #2 + dgm

Arithmetic shift Right — AShift R2, R2, #1 + dgm

1 X 3 = 3 M

(i) clear lower order 4 bits of R2 (16 bit)

AND R2, R2, 1111 0000

[OR]

AND R2, R2, #FFF0

(ii) Set msb of R2

OR R2, R2, 1000 - - - 0

[OR]

OR R2, R2, #8000

3(b) Dgm of stack indicating BOTTOM, TOP, TOS, SP. ——— (01M)

Explanation of stack ——— (01M)

	RISC	CISC
PUSH :	SUB SP, SP, #4 Store R2, (SP)	Move -(SP), R2
POP :	Load R2, (SP) Add SP, SP, #4	Move R2, (SP)+

— $1\frac{1}{2} \times 2 = 03M$